# Multi-Scale Drought Forecasting in Maharashtra using Integrated Remote Sensing, Climate Data, and Deep Learning Framework

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**ARTICLE HISTORY**

**ABSTRACT**

In Maharashtra, India, drought is a constant threat to farming, water security, and the stability of the economy and society. Conventional forecasting techniques frequently exhibit deficiencies in spatial specificity and predictive efficacy requisite for effective, localized mitigation. This study presents a district-level, multi-scale drought forecasting framework that amalgamates remote sensing, climate data, and both machine and deep learning models. The framework uses multi-source datasets from 2015 to 2024, such as CHIRPS rainfall, MODIS-derived NDVI and Land Surface Temperature, NASA FLDAS multi-depth soil moisture, and Evapotranspiration (ET). It also uses lagged temporal features and drought indices like the Standardized Precipitation Index (SPI) to capture complex temporal dependencies.

We compared the performance of Random Forest (RF), XGBoost, and Gradient Boosting to that of sequence-based models like LSTM and GRU.Deep learning models did better than tree-based ensembles.The LSTM model achieved the highest overall accuracy and demonstrated balanced performance across all drought categories, outperforming both Random Forest and XGBoost. Adding lagged variables made short-, medium-, and long-term forecasts much more accurate. The forecasts are shown on an interactive dashboard that gives farmers and policy makers useful information they can use to prepare for drought and better manage water resources. This work shows that using advanced machine learning with data from many sources can help build reliable, high-resolution drought early warning systems in sensitive agroclimatic areas.

**KEYWORDS**

Drought Forecasting; Remote Sensing; Machine Learning; Random Forest; LSTM; Soil Moisture; Maharashtra

## 1. Introduction

In Maharashtra, droughts are a common and severe problem that have a significant effect on water supplies, agriculture, and the socioeconomic well-being of millions of people (Kulkarni et al. 2020; Pande et al. 2024; "Meteorological Drought Evaluation" 2025). Due to the region's heavy reliance on the monsoon, crop failures, water shortages, and financial difficulties for farmers and communities can all result from interruptions in rainfall. In order to lessen these adverse effects, promote sustainable resource use, and facilitate improved planning at the local and regional levels, it is imperative to predict droughts precisely and as early as feasible (Nandgude et al. 2023). Over the years, conventional metrics such as the Standardized Precipitation Index (SPI) and the Standardized Precipitation Evapotranspiration Index (SPEI) have yielded valu-

able data for drought monitoring (Mahawadiwar et al. 2025; "Drought Monitoring and Prediction" 2022; "Meteorological Drought Evaluation" 2025). However, these indices typically rely on limited weather data and offer broad, coarse spatial resolution, which makes it difficult to capture localized variations in drought conditions that are critical for decision-making on the ground ("Agricultural Drought Monitoring" 2024; Zeynolabedin, Ehteshami, and Tamošaitienė 2022).

We now have more chances to enhance drought prediction systems because of developments in machine learning and satellite remote sensing (Gizachew Assefa and Terefe 2025; "Drought Prediction using Artificial Intelligence" 2024; Nandgude et al. 2023). In order to develop a more thorough and sophisticated understanding of drought, this study makes use of a variety of data sources, such as soil moisture profiles from NASA FLDAS ("Drought Prediction using Artificial Intelligence" 2024), vegetation health indicators like MODIS NDVI ("Exploration and Advancement" 2023; "Agricultural Drought Analysis" 2024), land surface temperature (LST) ("Agricultural Drought Analysis" 2024), and satellite-based rainfall estimates from CHIRPS. Reliability in predicting the onset and severity of drought is enhanced by the inclusion of lagged temporal features, which acknowledge that changes in vegetation health or soil moisture frequently take place some time after rainfall deficits (Watford, Bauch, and Anand 2024; Zhang et al. n.d.).

The majority of current drought monitoring methods concentrate on precipitation or precipitation plus evapotranspiration, frequently ignoring important information like soil moisture content at different depths and the actual state of the vegetation ("Drought Prediction using Artificial Intelligence" 2024; Anderson et al. 2018; Ekundayo et al. 2021). Their applicability for district-level planning, where customized solutions are required to address Maharashtra's varied agro-climatic zones, is limited by their lack of fine spatial granularity (Kulkarni et al. 2020; "Agricultural Drought Monitoring" 2024). Due to this disparity, local farmers and legislators find it difficult to obtain accurate and timely information that can inform crop selection, water management, and mitigation tactics.

By creating a thorough machine learning framework specifically designed for district-level drought prediction, this study tackles these issues (Pande et al. 2024). In order to more accurately depict the complexity of drought, it methodically combines various remote sensing and climatic datasets (Kulkarni et al. 2020; Zeynolabedin, Ehteshami, and Tamošaitienė 2022). In order to better understand how adding lagged variables increases forecasting accuracy, the study compares the performance of popular tree-based algorithms like Random Forest and XGBoost (Park, Kim, and Lee 2019; Tihi and Popov 2023; Qaisrani et al. 2025) with sequence-based deep learning models like Long Short-Term Memory (LSTM) networks (Mahawadiwar et al. 2025; Zhang et al. n.d.). An interactive visualization dashboard was developed to help convert the technical outputs into understandable, practical insights for farmers, water managers, and policymakers, facilitating quicker and better-informed decision-making. This ensures that the results are applicable and actionable.

This study makes a number of important contributions overall. First, it shows that combining data from many sources can greatly improve the accuracy and resolution of drought predictions, much more than using just one indicator ("Agricultural Drought Monitoring" 2024; Zeynolabedin, Ehteshami, and Tamošaitienė 2022). Second, it does a thorough comparison of machine learning and deep learning methods in the specific context of Maharashtra's districts, finding the best models for early warning of drought (Pande et al. 2024; Nandgude et al. 2023). Finally, it closes the gap between advanced computational predictions and real-world use by providing a visualization system that

is easy to use. All of these improvements work together to give Maharashtra's semi-arid areas a strong, data-driven way to get ready for and deal with the constant threat of drought (Kulkarni et al. 2020; "Exploration and Advancement" 2023). This will help make the state's agricultural and water systems more resilient.

## 2. Study Area and Data

### 2.1. *Study Region*

The western Indian state of Maharashtra is the main subject of the study. The third largest state in India, covering an area of more than 307,713 km$^2$, boasts a variety of agro-climatic zones, such as the humid coastal Konkan belt, the semi-arid and arid Vidarbha, Marathwada, and Deccan plateau, which represent a mix of coastal, plateau, and semi-arid landscapes. The state is predominantly an agrarian economy with heavy dependence on monsoon rainfall, and even minor fluctuations in rainfall patterns can cause severe water shortages, crop stress, and socio-economic instability (Mahawadiwar et al. 2025; "Meteorological Drought Evaluation" 2025; Pande et al. 2024). Moreover, 70

The areas most impacted by drought are Marathwada and Vidharbha, which leaves them vulnerable to regular shortages of rainfall, groundwater depletion, and crop loss (Kulkarni et al. 2020). Recurring droughts have significant impacts on rural livelihoods, irrigation demand, and water resource planning. For analytical consistency, the state was examined at the district level, comprising 36 administrative districts grouped into six divisions: Amravati, Aurangabad (Chhatrapati Sambhajinagar), Konkan, Nagpur, Nashik, and Pune. The district boundaries were obtained from the FAO/GAUL (2015, Level 2) dataset within Google Earth Engine (GEE), filtered to include only Maharashtra. This spatial resolution enables localized analysis and supports district-scale drought forecasting (Pande et al. 2024). With this spatial and climatic heterogeneity, Maharashtra makes it a suitable testbed for district-level, data-driven drought prediction models. The variation in rainfall patterns, soil moisture levels, and vegetation response makes it extremely valuable and diverse for model training and validation (Kulkarni et al. 2020; Mahawadiwar et al. 2025; Pande et al. 2024).

### 2.2. *Data Sources*

The following study integrates numerous remote sensing and climatic datasets to capture the multifaceted nature of drought in Maharashtra, India ("Agricultural Drought Monitoring" 2024; Kulkarni et al. 2020; Zeynolabedin, Ehteshami, and Tamošaitienė 2022). This extensive study spans a decade, from January 2015 to December 2024, and examines the dynamics of rainfall, vegetation health, surface temperature, soil moisture, and evapotranspiration.

#### *2.2.1. Rainfall*

Rainfall data was sourced from the Climate Hazards Group InfraRed Precipitation with Station data (CHIRPS) dataset. This product integrates satellite-based infrared precipitation estimates with in-situ station data to create a high-resolution, bias-corrected rainfall record. Daily precipitation values were spatially averaged for each district and aggregated into monthly totals. These monthly totals were then used to com-

pute key drought indices, including the Standardized Precipitation Index (SPI) and the Standardized Precipitation Evapotranspiration Index (SPEI), which are widely used for monitoring meteorological drought conditions ("Drought Monitoring and Prediction" 2022; Mahawadiwar et al. 2025; "Meteorological Drought Evaluation" 2025).

#### *2.2.2. Vegetation Health*

Vegetation health was monitored using the Moderate Resolution Imaging Spectroradiometer (MODIS) Normalized Difference Vegetation Index (NDVI) from the MOD13A3 Aqua dataset. Monthly averages of NDVI values were computed for each district and scaled by the required factor of 0.0001. NDVI serves as a crucial proxy for vegetation stress, as a decline in "greenness" often provides an early warning signal of agricultural drought, sometimes before other indicators become apparent ("Agricultural Drought Analysis" 2024; Anderson et al. 2018; "Exploration and Advancement" 2023).

#### *2.2.3. Surface Temperature*

Land Surface Temperature (LST) patterns were derived from the MODIS LST data product (MOD11A2). Raw temperature readings were converted to Celsius using the formula (LSTraw$\times$0.02)–273.15 and then averaged monthly for each district. LST is a critical variable for understanding heat accumulation and its influence on evapotranspiration in response to changing moisture conditions ("Agricultural Drought Analysis" 2024; Kulkarni et al. 2020).

#### *2.2.4. Soil Moisture*

Soil moisture data was obtained from the NASA Famine Early Warning Systems Network (FEWS NET) Land Data Assimilation System (FLDAS). The analysis included four distinct depth layers (0–10 cm, 10–40 cm, 40–100 cm, and 100–200 cm), providing a comprehensive vertical profile of soil water storage. This dataset was essential for assessing subsurface moisture variability and its impact on agricultural drought ("Drought Prediction using Artificial Intelligence" 2024).

#### *2.2.5. Evapotranspiration*

Evapotranspiration (ET) data was sourced from the MODIS product MOD16A2. The raw values were converted to millimeters by applying a scale factor of 0.1 and then averaged monthly for each district. As ET represents the combined water loss from surface evaporation and plant transpiration, it is a vital indicator of water stress and drought intensity ("Drought Monitoring and Prediction" 2022).

All datasets were processed within the Google Earth Engine (GEE) platform using its Python API to ensure computational efficiency for large-scale spatiotemporal analysis. The processed variables were then consolidated into a unified dataset in CSV format for the subsequent drought modeling steps.

### 2.3. *Data Preprocessing*

All datasets were subjected to a thorough preprocessing workflow prior to modeling in order to guarantee their accuracy, consistency, and integration readiness. Standardization was a crucial step in preserving coherence across variables and timeframes because

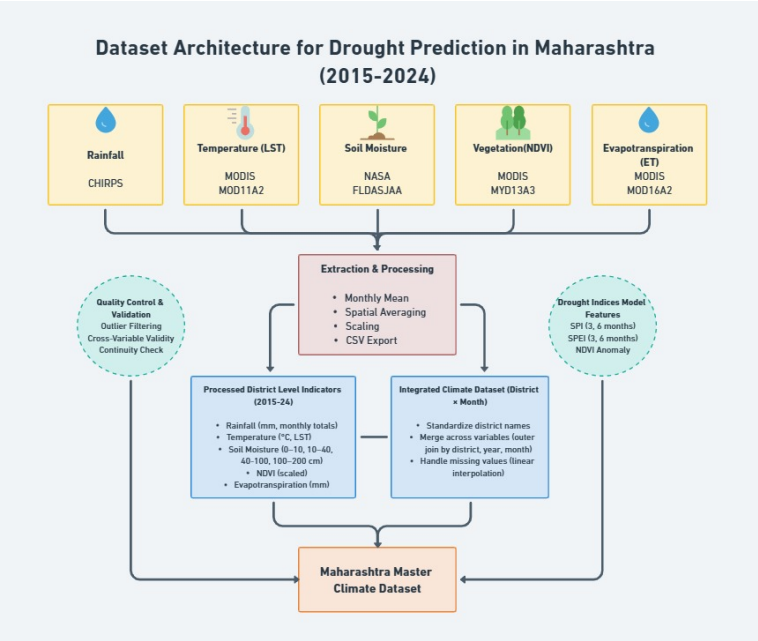


Figure 1.: Workflow for integrating multi-source remote sensing data for the drought prediction model.

the data came from various satellite sensors and sources, each of which had its own temporal frequency, resolution, and units (”Agricultural Drought Monitoring” 2024; Kulkarni et al. 2020; Zeynolabedin, Ehteshami, and Tamošaitienė 2022).

Initially, every dataset underwent a monthly temporal resolution resampling. Google Earth Engine (GEE) was used to compute monthly averages (or totals, if applicable) for daily or 8-day products such as CHIRPS rainfall, MODIS LST, and MODIS ET. This made sure that from January 2015 to December 2024, each variable had a consistent time step.

Then we aggregated the data at the district level using the FAO GAUL Level 2 boundary shapefile. This then was undergone through a clean spatial averaging which was performed on all pixels within each district boundary resulting in one representative value per district per month for each variable. This was extremely important for aligning the project’s goal of district-level drought prediction, similar to other regional studies in Maharashtra (Pande et al. 2024), rather than as opposed to pixel-level or regional analyses.

To handle missing values, a combination of temporal interpolation and spatial smoothing was used. Missing entries caused by cloud contamination (in MODIS NDVI or LST) or data gaps in CHIRPS were filled using linear interpolation along the time dimension. For isolated missing districts, neighbor-based averaging was applied to maintain spatial consistency.

After all variables were normalized, scaling and unit conversions were applied to standardize. MODIS NDVI values were rescaled by $\times 0.0001$, and LST was converted from Kelvin to Celsius through $(\text{LSTraw} \times 0.02) - 273.15$. ET values were multiplied by 0.1 in order to convert them to millimeters.

Lagged variables of 1-3 months were established for rainfall, soil moisture, NDVI, and LST to account for delayed ecosystem responses, as vegetation tends to respond weeks following rainfall deficiencies (Mahawadiwar et al. 2025; Watford, Bauch, and

Anand 2024; Zhang et al. n.d.). Then, all datasets were joined into one dataframe based on the district and month-year keys. The final dataset was normalized for model training purposes and contained monthly climatic and biophysical conditions for every district, creating a robust, multi-faceted dataset for predictive modeling (Kulkarni et al. 2020; Zeynolabedin, Ehteshami, and Tamošaitienė 2022).

### 2.4. *Derived Variables*

We have certain derived variables which eventually helped us in making our dataset ready for model training and validation purposes. To enhance the model accuracy, we used drought indices and added derived features which were computed from the raw datasets. The Standardized Precipitation Index (SPI) and Standardized Precipitation Evapotranspiration Index (SPEI) were calculated at 3-month and 6-month scales to represent the overall short-term and medium-term moisture conditions ("Drought Monitoring and Prediction" 2022; Mahawadiwar et al. 2025; "Meteorological Drought Evaluation" 2025).

SPI is computed as follows ("Meteorological Drought Evaluation" 2025; Rezaiy and Shabri 2025):

$$\text{SPI} = \frac{P_i - \overline{P}}{\sigma_P} \tag{1}$$

where $P_i$ is the precipitation for a given period, $\overline{P}$ is the long-term mean, and $\sigma_P$ is the standard deviation of rainfall.

This was further extended by another variable SPEI, which also accounts for evapotranspiration and is defined as follows ("Drought Monitoring and Prediction" 2022):

$$\text{SPEI} = \frac{(P_i - ET_i) - (\overline{P} - \overline{ET})}{\sigma_{(P-ET)}} \tag{2}$$

We also used NDVI anomalies to quantify attributes such as Vegetation Stress from multiple years of monthly means ("Agricultural Drought Analysis" 2024; Anderson et al. 2018). Negative anomalies signal reduced vegetation vigor, while positive values reflect healthy growth conditions. To further represent the delayed ecosystem results, lagged features (1–3 months) of rainfall, soil moisture, NDVI, and land surface temperature were added, making it easier to recognize how drought effects unfold over time (Mahawadiwar et al. 2025; Watford, Bauch, and Anand 2024; Zhang et al. n.d.). Together, these variables create a balanced dataset that captures both climatic variability and environmental feedback across Maharashtra's districts (Kulkarni et al. 2020; Zeynolabedin, Ehteshami, and Tamošaitienė 2022).

## 3. Literature Survey

Drought is a devastating natural hazard with consequential impacts on ecosystems, agriculture, and society. Provided its slow onset and complex nature, the development of robust monitoring, forecasting, and early warning systems is essential to mitigate

its adverse effects (Nandgude et al. 2023; Tihi and Popov 2023). The research landscape has evolved from traditional statistical methods to advanced remote sensing and machine learning approaches to better predict drought events and understand their effects on a deeper level (Nandgude et al. 2023; Watford, Bauch, and Anand 2024).

### 3.1. *From Single-Indicator to Combined Drought Monitoring*

Historically, drought monitoring has relied on meteorological indices calculated from ground data. The Standardized Precipitation Index (SPI) is one of the most widely used indices due to its simplicity, relying solely on precipitation data to quantify deficits over time (Mahawadiwar et al. 2025; "Meteorological Drought Evaluation" 2025; Pande et al. 2024). However, a singular focus on rain can be insufficient to capture the full scope of agricultural drought, which also involves factors like soil moisture deficits and vegetation health ("Agricultural Drought Analysis" 2024; "Agricultural Drought Monitoring" 2024; Kulkarni et al. 2020).

To overcome this limitation, researchers have increasingly used remote sensing data which offers synoptic-scale monitoring capabilities (Anderson et al. 2018; "Exploration and Advancement" 2023). Vegetation indices such as the Normalized Difference Vegetation Index (NDVI) and the Normalized Difference Water Index (NDWI) are commonly used to assess vegetation health and water stress using satellite imagery ("Agricultural Drought Analysis" 2024; Ekundayo et al. 2021). Building on this, studies have developed composite indices like the Normalized Difference Drought Index (NDDI), which is derived from NDVI and NDWI to provide a more precise measure of drought severity ("Exploration and Advancement" 2023).

Realizing that agricultural drought is a composite effect of unfavorable vegetation, water stress, and low soil moisture, much has been the trend towards the creation of Combined Drought Indicators (CDIs) ("Agricultural Drought Monitoring" 2024; Kulkarni et al. 2020; Zeynolabedin, Ehteshami, and Tamošaitienė 2022). These composite indices combine several variables to give a more integrated evaluation. For example, a regional CDI was generated by merging remotely sensed information on vegetation health, soil moisture, and crop water stress with a clear weighting approach ("Agricultural Drought Monitoring" 2024). Likewise, a CDI for the Marathwada region of India was built by merging SPI, Land Surface Temperature (LST), Soil Moisture (SM), and NDVI, where Principal Component Analysis (PCA) was utilized to assign weights to the input parameters (Kulkarni et al. 2020).

### 3.2. *Advancements in Drought Forecasting with Machine Learning*

Whereas monitoring gives the present picture, forecasting is needed for effective drought management. Predicting using machine learning (ML) and artificial intelligence (AI) models has become increasingly prominent in the field (Min 2025; Nandgude et al. 2023; "Drought Prediction using Artificial Intelligence" 2024). Data-driven methods are usually better at extracting the intricacies of the non-linear relationships in climatic variables than are the older stochastic methods such as the Autoregressive Integrated Moving Average (ARIMA) (Tihi and Popov 2023).

A popular approach is to train ML models to predict drought indices for different lead times ("Drought Monitoring and Prediction" 2022; "Meteorological Drought Severity Forecasting" 2025). A model involving Support Vector Machines (SVM) and Boosted Trees was used in a study at Nashik, Maharashtra, to predict SPI with a high

coefficient of determination ("Forecasting of Meteorological Drought" 2024; Pande et al. 2024). Another research created a set of AI-drought indices with models such as Random Forest (RF) and proved that these AI-drought indices performed better than nine traditional indices when compared to drought indicators such as soil moisture and runoff ("Drought Prediction using Artificial Intelligence" 2024). Likewise, a study in Ethiopia established Random Forest and Gradient Boosting models as efficient predictors for the Palmer Drought Severity Index (Gizachew Assefa and Terefe 2025).

### 3.3. *Methodological Innovations for Enhanced Prediction*

To better enhance the accuracy of forecasting, scholars are investigating new data processing methods and hybrid models. Time series in hydrology tend to be non-stationary, which may contradict the performance of typical ML models. To resolve this, a hybrid model combining Discrete Wavelet Transform (DWT) with SVM (W-SVM) was introduced to decompose and predict the SPI time series, performing better than isolated models (Rezaiy and Shabri 2025).

Some of the other novel strategies are modifying the input parameters themselves. A study constructed a Severe Drought Area Prediction (SDAP) model based on Random Forest that forecast soil moisture in the future without employing meteorological information as a direct variable (Park, Kim, and Lee 2019; Qaisrani et al. 2025). It uses land surface variables like topography, vegetation, and thermal information to forecast areas with high agricultural drought risk under non-rainfall conditions (Park, Kim, and Lee 2019).

Recently, deeper architectures of deep learning for spatio-temporal data have been utilized in drought forecasting. Architectures such as Convolutional LSTM (ConvLSTM) and transformer-based architectures have been utilized to predict drought indices 12 months in advance ("Long-Term Drought Prediction" 2023). Long Short-Term Memory (LSTM) networks have also been demonstrated in other research with potential applications to drought forecasting using SPI and data such as MERRA-Land (Mahawadiwar et al. 2025; Zhang et al. n.d.). This points to the capability of deep learning to address the complexities of long-range drought forecasting ("Long-Term Drought Prediction" 2023; Zhang et al. n.d.).

## 4. Methodology

### 4.1. *Overview of the Workflow*

The workflow of this study follows a structured, multi-stage process beginning with the data collection process and ending with visual drought mapping. The process begins with the acquisition of multi-source datasets which comprises both climatic and environmental factors which are relevant to the prediction model of drought in Maharashtra (Kulkarni et al. 2020; Zeynolabedin, Ehteshami, and Tamošaitienė 2022). This includes satellite-based rainfall data from CHIRPS, vegetation indices such as NDVI and land surface temperature data from MODIS ("Agricultural Drought Analysis" 2024), along with soil moisture profiles collected from NASA's FLDAS databases ("Drought Prediction using Artificial Intelligence" 2024). The dataset was collected from the time interval of 2015 to 2024 and which was further aggregated at the district level so as to align with our geographic scope.

To ensure both spatial and temporal consistency, a number of data preprocessing

steps were held including cleaning, identifying missing values, resampling and importantly feature alignment to create a unified dataset. Subsequently, feature engineering was done to start with lagged temporal variables which can capture the delayed vegetation and climatic responses to map with rainfall deficiency which is in fact, an essential aspect of drought prediction (Watford, Bauch, and Anand 2024; Zhang et al. n.d.). The SPEI-based drought classification was then further applied to label the particular district and month record into 3 categorical drought classes, which provided the ground truth values of the model ("Drought Monitoring and Prediction" 2022).

Now after the dataset was prepared accordingly, three different machine learning algorithms were employed to predict the drought-labelled classes using the supervised learning methods. These algorithms included Random Forest (Park, Kim, and Lee 2019; Tihi and Popov 2023), XGBoost (Gizachew Assefa and Terefe 2025), and LSTM (Mahawadiwar et al. 2025; Zhang et al. n.d.), which would be further discussed in detail in the subsequent sections. Every model was trained and validated using a time-series split in order to preserve the chronological order. Then these trained models were evaluated based on the performance metrics such as accuracy, precision, recall and the F-1 score. Finally, an interactive visualisation dashboard was also made to translate the model predictions outputs into color-coded drought maps and trend graphs which provided an accessible and actionable interface to stakeholders, policymakers, water managers and farmers. This overall integrated workflow ensured an end-to-end approach starting from raw data to real world prediction model aimed at enhancing drought preparedness and management across the state of Maharashtra.

### 4.2. *Feature Engineering*

In order to enhance the predictive performance of the model, we employed various feature engineering techniques. Since drought is inherently a temporal and a multidimensional phenomenon, all these raw climatic and environmental datasets were transformed to better capture both short-term and long-term variability. Lagged features were made to generate key attributes such as rainfall, soil moisture, NDVI and land surface temperature which in turn would represent the delayed influence of rainfall and temperature changes affecting the soil and vegetation eventually. This particularly helped with the LSTM network, since lags of one, two and three months were introduced, making it possible to efficiently learn about the temporal dependencies and evolving drought patterns (Mahawadiwar et al. 2025; Watford, Bauch, and Anand 2024). Furthermore, to preserve the cyclical nature of temporal variables such as month, sine and cosine transformations were applied to encode seasonality effectively. This ensured that months which were temporally adjacent (such as January and December) are represented closely in the feature space and lie in the same probabilistic window. These sin-cos embeddings would better capture the seasonal dependencies and make it uniform and valid yearly.

After lag generation, scaling and normalization was employed so that all the input variables lie in a comparable range. This step was essential because the dataset collected was from diverse perspectives and included different sources and units (e.g., rainfall in millimeters, NDVI as a ratio, and temperature in Celsius). It was constrained in between 0 and 1 ensuring balanced model training and preventing features with large magnitudes dominating the learning process. Additionally, categorical encoding was used where necessary encoding was needed to represent district-level identifiers and temporal variables such as moth or season. The drought classification labels

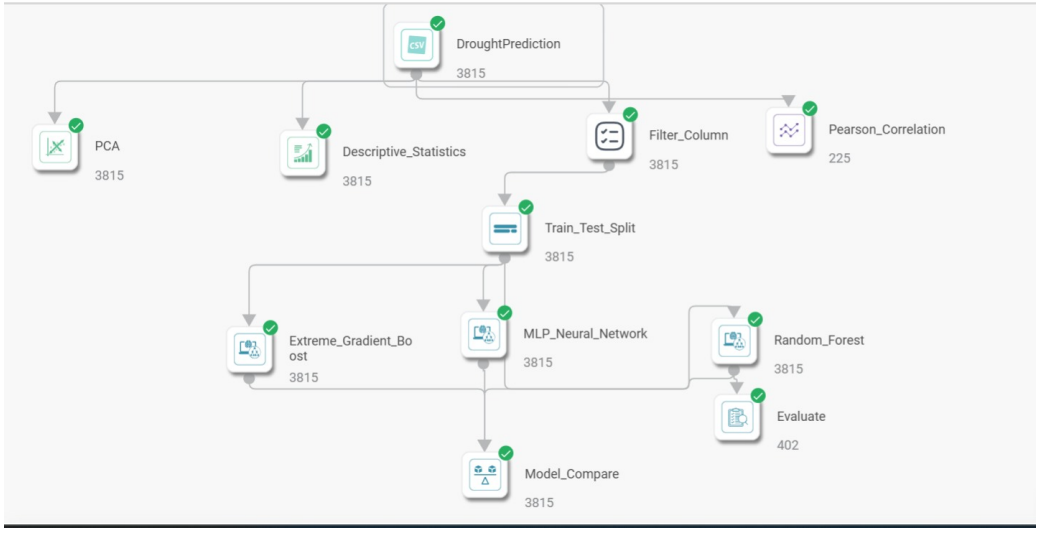


Figure 2.: Advanced workflow incorporating SMOTE for class imbalance and a parallel cross-validation pipeline.

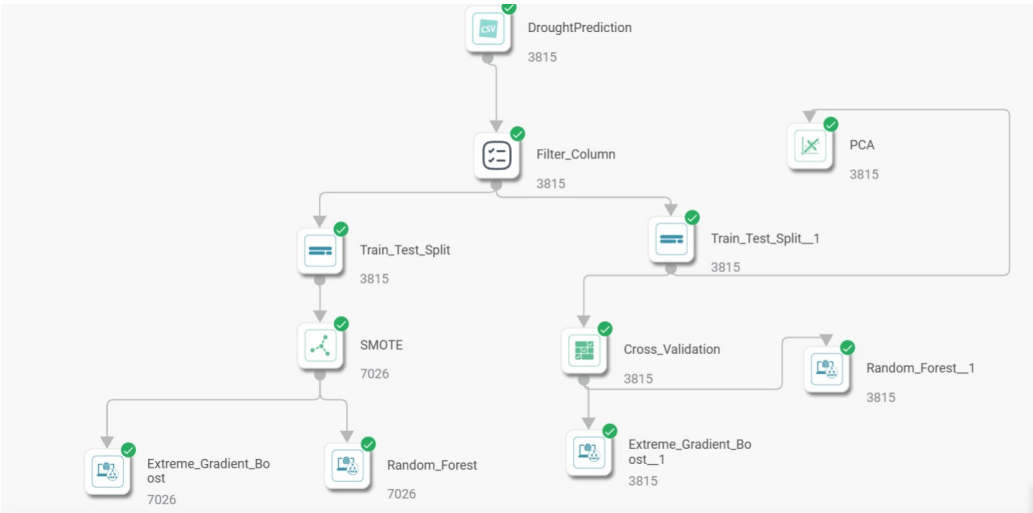


Figure 3.: Workflow for drought prediction model development and comparison on the Rubyscape platform.

were derived using the Standardized Precipitation Evapotranspiration Index (SPEI) ("Drought Monitoring and Prediction" 2022). Based on the established drought value threshold observations with SPEI 1.5 were classified as Extreme Drought, those with SPEI 1 as Moderate Drought, and all others as No Drought. This formed the target classes for supervised learning aligning all subsequent modeling efforts towards accurate drought class prediction. This provided a robust and unique foundation for the training and validation steps for the machine learning and deep learning models on how the drought class might be predicted based on different features, values, trends and dependencies.

### 4.3. *Model Development*

This study implemented three machine learning models, Random Forest (RF), Extreme Gradient Boosting (XGB) and Long Short-Term Memory (LSTM). All these models were used to predict drought conditions class labels across Maharashtra's districts. Each model was trained on the preprocessed and feature engineering dataset and evaluated using consistent validation criteria. For the initial baseline, Random Forest (RF) was employed on the training data. It used ensemble learning to reduce the overall variance and eventually improve prediction stability, a technique used in several drought studies (Park, Kim, and Lee 2019; Qaisrani et al. 2025; Tihi and Popov 2023). Different Models of RF alone was employed which included Baseline RF, SMOTE RF (which handled the problem of class imbalance through Synthetic Minority Oversampling), and Hyperparameter-Tuned RF. Among them, the SMOTE-tuned model turned out to be the best performing model, achieving around 85

Finally, the most advanced model, Long Short-Term Memory (LSTM), was developed to capture the sequential dependencies in the climatic and agricultural features (Mahawadiwar et al. 2025; Zhang et al. n.d.). LSTM networks are a specialized form of recurrent neural networks, which are capable of learning the temporal relationships through the approach of gated mechanisms, by keeping the relevant past information too. This study further enhanced when a district-wise training was employed, this included training each of the 34 districts individually through separate LSTM models. This approach allows the model to learn the intrinsic features catered to each particular district, a concept aligned with studies focusing on specific regions like Nashik (Pande et al. 2024). In the same model, temporal features such as NDVI, soil moisture, and temperature were used explicitly as sequential inputs, which garnered the core concept of the LSTM model. This resulted in gaining the best accuracy of around 92

The models were trained on a time series cross validation method to make sure it maintains the temporal integrity between training and testing splits (Tihi and Popov 2023). Also for LSTM, an Adam optimizer was used with a learning rate tuned through experimentation while early stopping was employed to prevent overfitting on the data. Overall the LSTM's performance and adaptability established it to be the most reliable model for drought prediction within this framework.

### 4.4. *Evaluation Metrics and Validation Accuracy*

In order to evaluate the model performance, various evaluation metrics such as Accuracy, Precision, Recall, F1 score were used to understand how effectively each approach identified the drought and non-drought conditions classes. To reflect the overall cor-

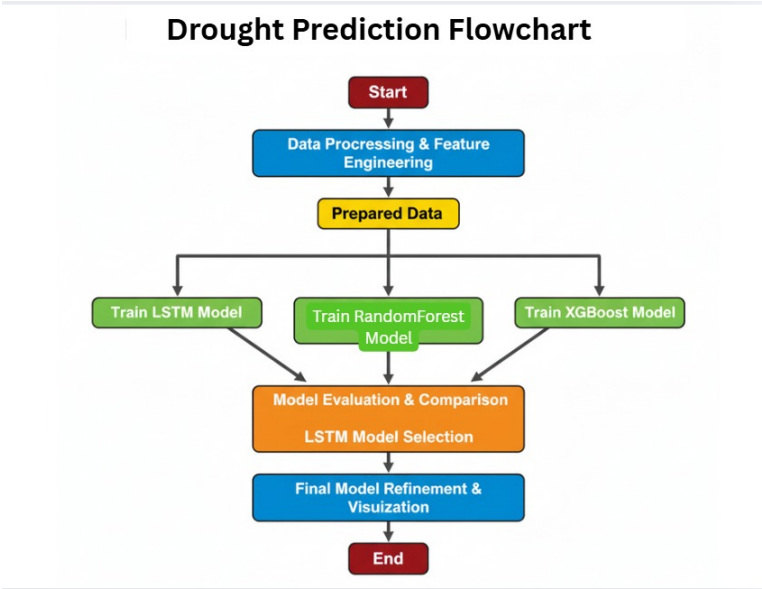


Figure 4.: Flowchart of the drought prediction methodology, from data preparation to final model selection.

rectness, we used accuracy, and on the other hand, precision and recall helped us assess how well the models detected true-drought events without too many false alarms. Furthermore, the F-1 Score which is the harmonic mean of Precision and Recall gave us a balanced measure between both, capturing the extreme cases and the drought events which were less frequent. In addition to this, to visualize prediction errors and better interpret the model's prediction classification behavior, a Confusion Matrix was used.

To maintain the natural time order of climatic data, a time-series cross-validation method was adopted for making the prediction instead of random shuffling. This ensured that past data was always used to predict future data conditions, learning in a future-oriented way while subsequently preserving real-world forecasting logic (Mahawadiwar et al. 2025; Tihi and Popov 2023). Each fold tested the models on unseen future segments and the results gained were averaged to make sure the consistency holds. Among all, the LSTM model consistently achieved the best and highest achieving accuracy and stability too, which highlighted its strong and grasping ability to capture the sequential dependencies and relationships in Maharashtra's climatic data (Zhang et al. n.d.).

## 5. Results and Discussions

### 5.1. *Model Performance*

The predictive performance results of the study model are summarized and illustrated in Figures 5-7. The LSTM models clearly outperform both Random Forest and XGBoost across overall accuracy and also the class-wise F1 scores, achieving an accuracy of 92.1

Class-wise metrics emphasize LSTM's balanced approach towards all categories of drought classes. For extreme drought, LSTM achieves an F1-score of 0.84, comparable to Random Forest (RF) and XGBoost. For moderate drought, however, LSTM

improves predictions substantially with an F1-score of 0.31 compared to tree-based models (RF: 0.29, XGB: 0.34). This clearly suggests its superior and much more accountable capability in learning temporal patterns and mitigating class imbalance, which is a key advantage of sequential models (Mahawadiwar et al. 2025; Zhang et al. n.d.).

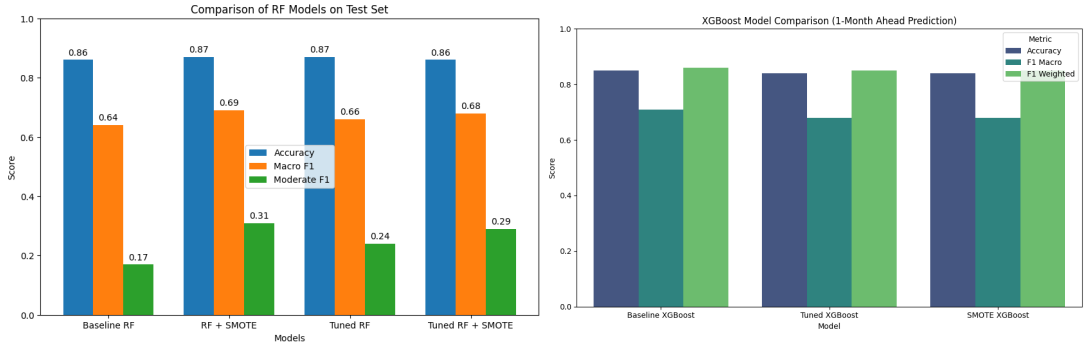


Figure 5.: (a) The application of SMOTE significantly enhances the RF model's ability to identify the 'Moderate' drought class. (b) Comparison of Baseline, Tuned, and SMOTE-applied XGBoost models for 1-month ahead prediction.

The comparison bar chart (Figure 6) visually confirms LSTM's dominance over the other two models, and the detailed classification tables provide a clear breakdown of class-specific strengths and limitations. These results indicate that for operational drought forecasting where class balance is crucial, the LSTM model, whose core principle is working on sequential dependencies, offers the best trade-off between overall accuracy and trustworthy detection across each drought class label. This aligns with findings that more advanced or hybrid models often yield superior and more robust performance compared to isolated or simpler models ("Long-Term Drought Prediction" 2023; Rezaiy and Shabri 2025). In contrast, RF and XGBoost may be considered complementary models where interpretability or quick deployment are the primary goals.

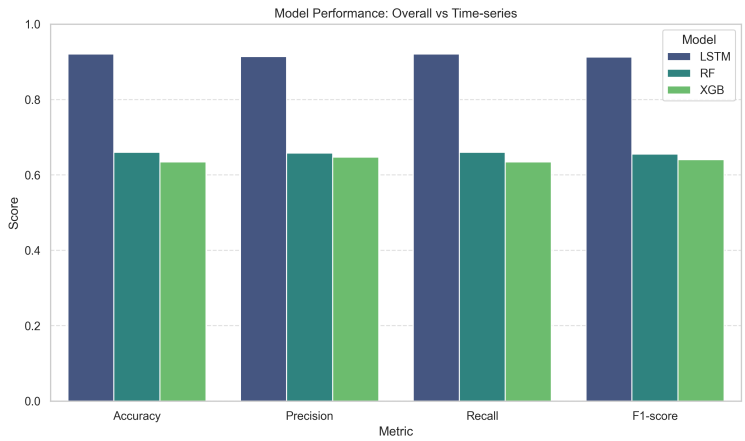


Figure 6.: Comparative metrics of the LSTM, Random Forest (RF), and XGBoost (XGB) models on the time-series test set. The results demonstrate the superior performance of the LSTM model in overall drought prediction accuracy.

### 5.2. *Spatial and Temporal Insights*

Effective management of drought requires a close understanding of both where a drought is occurring (spatial dimension) and how this is varying over time (temporal dimension). Much effort is seen in the literature to progress models to provide high-resolution spatial data with extended temporal forecasting horizons.

#### *5.2.1. Spatial Dimensions and High-Resolution Mapping*

One of the key trends in drought research is the move from general, large-scale assessment to high-resolution, localized analysis that can inform on-the-ground decision-making. Research focuses increasingly on specific administrative and agro-climatic zones, such as the Marathwada region (Kulkarni et al. 2020), Nashik district ("Forecasting of Meteorological Drought" 2024; Pande et al. 2024), and Parbhani ("Meteorological Drought Evaluation" 2025) in Maharashtra, indicating a need for tailored, sub-regional models. This specificity is used globally, with research targeting exact locations like Ethiopia (Gizachew Assefa and Terefe 2025), Nigeria's Sudano-Sahelian region (Ekundayo et al. 2021), and northeast Thailand (Qaisrani et al. 2025).

It heavily relies on geospatial technology and satellite remote sensing (Akbari and Torabi Haghighi 2022; Anderson et al. 2018; "Exploration and Advancement" 2023). High-level models now incorporate land surface parameters, such as topography and vegetation data, to predict severe drought areas without direct meteorological inputs, which is a more realistic tool for water allocation (Park, Kim, and Lee 2019). The objective is to move from simple monitoring to proactive spatial forecasting, using algorithms like Random Forest and Analytic Hierarchy Process to represent drought vulnerability at a more refined resolution (Qaisrani et al. 2025). Global initiatives like the Global Drought Observatory (GDO) similarly emphasize requirements for high-resolution spatial data, for instance in evaluating modelled daily discharge for operational monitoring (Cammalleri, Barbosa, and Vogt 2020).

#### *5.2.2. Temporal Dynamics and Forecasting Horizons*

The temporal scale of drought science has evolved from historical analysis to sophisticated forecasting on greater than one time scale. While traditional time-series models like ARIMA remain used, they are increasingly outcompeted by machine learning models like Random Forest that manage sophisticated, non-linear intertemporal relationships in climate data (Tihi and Popov 2023).

A first objective is to increase the lead time for predictability. Most studies concentrate on short- to medium-range forecasting of indices like the SPI, optimally for 3- to 6-month lead times ("Forecasting of Meteorological Drought" 2024; Pande et al. 2024). The frontier of research, however, is moving towards precise long-term forecasting a year in advance, which is most critical for strategic farm planning ("Long-Term Drought Prediction" 2023). This has been done by employing deep learning models such as Long Short-Term Memory (LSTM) and Transformer models that are able to process complex spatio-temporal information (Mahawadiwar et al. 2025; "Long-Term Drought Prediction" 2023; Zhang et al. n.d.).

To achieve improved accuracy, scientists are also making use of novel methods of processing data. For instance, Discrete Wavelet Transforms (DWT) are used to decompose non-stationary time-series data into relatively more stable and predictable components prior to their input into forecasting models like Support Vector Machines (SVM) (Rezaiy and Shabri 2025). Besides that, new interest is being generated for

methods inspired by dynamical systems to better pick up on the temporal mechanics behind drought development and divert from purely data-driven approaches (Watford, Bauch, and Anand 2024).

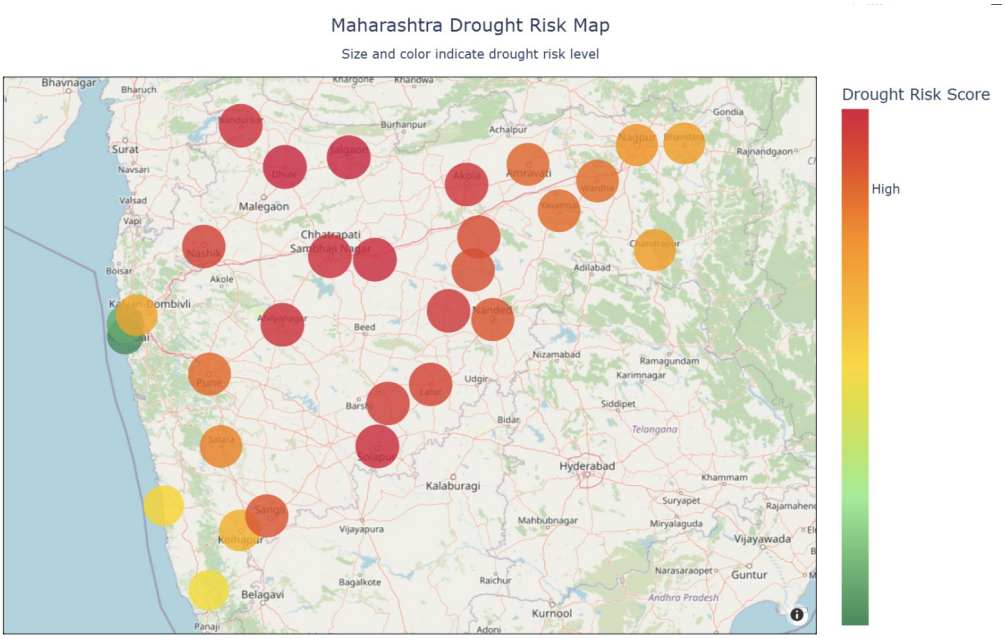


Figure 7.: Drought Risk Score's spatial distribution throughout Maharashtra's districts. Both size and color are used to indicate the degree of risk; low-risk areas are represented by green markers, while high-risk districts are represented by red markers.

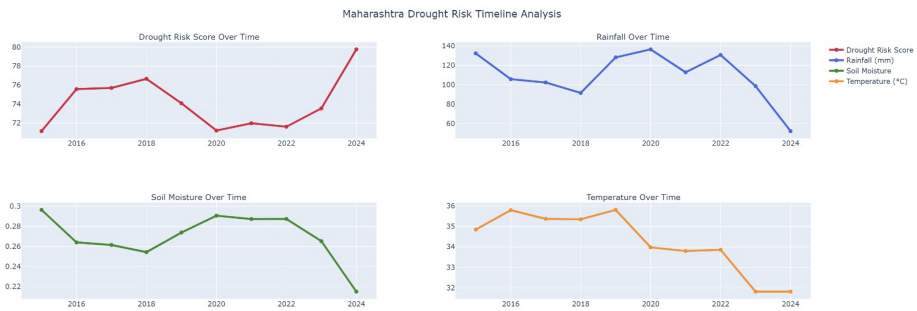


Figure 8.: Timeline analysis of Drought Risk Score alongside key climatic drivers in Maharashtra (2016–2024).

### 5.3. *Feature Importance and Interpretation*

The foundation of any machine learning model is feature analysis, which gives us a better understanding of the function of each component and attribute in the input datasets. It makes us draw attention to the drought prediction models' primary and main cause drivers. The study showed that the most significant characteristics are

consistently found to be rainfall, NDVI anomalies, and soil moisture (Mahawadiwar et al. 2025; "Agricultural Drought Analysis" 2024; "Drought Prediction using Artificial Intelligence" 2024). Additionally, seasonal embeddings and temporal lag features also improve the LSTM's ability to detect early warning signals, which clearly identifies its sequential dependencies (Mahawadiwar et al. 2025; Zhang et al. n.d.). Furthermore, rainfall deficits are shown to be the primary cause of extreme drought ("Meteorological Drought Evaluation" 2025), while moderate drought led to sensitivity to both soil moisture patterns and the combined NDVI ("Agricultural Drought Analysis" 2024; Kulkarni et al. 2020). These features have different and independent effects on each of the drought class labels and hence makes it extremely important to handle each of these features and their influence on the various sections of the model. These revelations highlight the multi-source datasets' general applicability. These insights underscore the overall relevance of multi-source datasets and engineered temporal features in improving drought detection accuracy for the corresponding predictive model (Nandgude et al. 2023; Zeynolabedin, Ehteshami, and Tamošaitienė 2022).

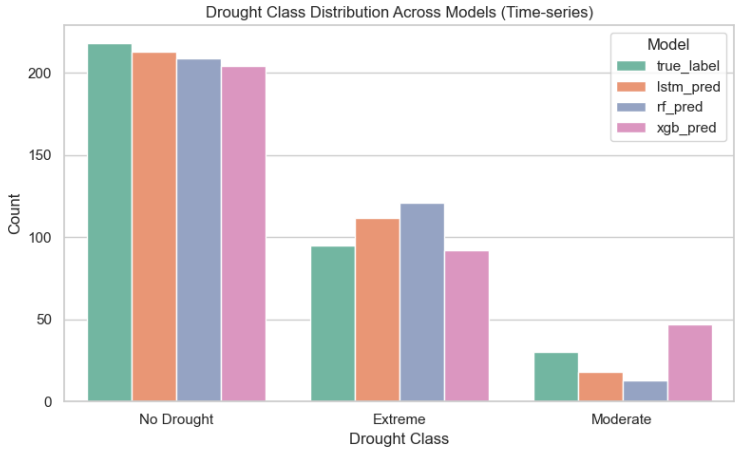


Figure 9.: Distribution of true vs. predicted drought classes for the LSTM, RF, and XGBoost models.

## 6. Future Work and Conclusion

After the research of this study, it is clear that for predicting district-level droughts in Maharashtra, conventional methods that rely on a single data source or large spatial scales often fall short in performance. This study proposed a novel machine learning framework that incorporated multi-source remote sensing datasets to better understand the complex relationships of climatic and ecological drought factors, leading to better accountability. Rainfall, vegetation health indicators (NDVI and LST), and multi-layer soil moisture levels were gathered from CHIRPS, MODIS, and NASA FLDAS. These were blended to mimic the interconnected nature of drought and accounted for lagged features, which capture the delayed response to rainfall shortfalls [15, 16]. These methods proved extremely important in predicting drought onset by providing the model with knowledge of how long it takes for soil moisture to deplete and crops to become stressed after a rainfall deficit begins.

When compared, tree-based models such as Random Forest and XGBoost showed

consistently good results for most districts but had a limited capacity to model sequential dependencies. This gap was filled by the LSTM-based deep learning model, which outperformed them when it came to grasping the complexities of non-linear temporal patterns. This highlights the significant potential of deep learning to advance how we monitor and predict droughtS. Furthermore, for policymakers, farmers, and other stakeholders, the visualization interface provides a tool for quick decision-making, allowing them to manage resources and plan interventions in drought-prone areas by taking timely actions.

Looking ahead, this framework can be further refined and expanded. The incorporation of transformer-based architectures could enhance the modeling of long-term dependencies, building on the success of deep learning in this field. Higher temporal resolution, such as weekly or bi-weekly predictions, could enable more rapid early warning systems. Future work may also succeed in integrating other dependencies such as groundwater levels, crop yield statistics, and socio-economic indicators to create a more holistic prediction system. This adaptive drought management tool brings us closer to a vision of climate-resilient agriculture in Maharashtra, demonstrating how the fusion of satellite data and artificial intelligence can become a critical tool for sustainable resource planning in one of India's most drought-prone regions.

## References


Akbari, M., and A. Torabi Haghighi. 2022. "Satellite-based Agricultural Water Consumption Assessment in the Ungauged and Transboundary Helmand Basin between Iran and Afghanistan." *Remote Sensing Letters* 13 (12): 1236–1248.

Anderson, M. C., C. A. Zolin, P. C. Sentelhas, et al. 2018. "A Machine-learning Approach to Forecasting Remotely Sensed Vegetation Health." *International Journal of Remote Sensing* 39 (6): 1635–1650.

Anonymous. 2022. "Drought Monitoring and Prediction using SPI, SPEI, and Random Forest." *IWA Publishing.*

Anonymous. 2023. "Exploration and Advancement of NDDI Leveraging NDVI and NDWI in Indian Semi-arid Regions: A Remote Sensing-based Study." *ScienceDirect.*

Anonymous. 2024a. "Agricultural Drought Analysis using NDVI and Land Surface Temperature." *ScienceDirect.*

Anonymous. 2024b. "Agricultural Drought Monitoring and Early Warning at the Regional Scale using a Remote Sensing-based Combined Index." *Environmental Monitoring and Assessment.*

Anonymous. 2024c. "Forecasting of Meteorological Drought using Ensemble and Machine Learning Models." *Environmental Sciences Europe.*

Anonymous. 2025a. "Meteorological Drought Evaluation in Parbhani, Maharashtra, using SPI." *Agronomy Journals.*

Anonymous. 2025b. "Meteorological Drought Severity Forecasting Utilizing Blended Ensemble Approach." *PMC Article* (PMC12268929).

Cammalleri, C., P. Barbosa, and J. V. Vogt. 2020. "Evaluating Simulated Daily Discharge for Operational Hydrological Drought Monitoring in the Global Drought Observatory (GDO)." *Hydrological Sciences Journal* 65 (8): 1316–1325.

Ekundayo, O. Y., E. C. Okogbue, F. O. Akinluyi, A. M. Kalumba, and I. R. Orimoloye. 2021. "Spatiotemporal Drought Assessment using Vegetation Health Index and Standardized Precipitation Index over Sudano-Sahelian Region of Nigeria." *African Geographical Review* 40 (4): 412–424.

Gizachew Assefa, T. M., and B. Terefe. 2025. "Machine Learning-based Drought Prediction using Palmer Drought Severity Index and TerraClimate Data in Ethiopia." *PMC Article*

(PMC12176240).

Kulkarni, S., et al. 2020. "Developing a Remote Sensing-based Combined Drought Indicator Approach for Agricultural Drought Monitoring over Marathwada, India." *MDPI*.

Mahawadiwar, V. N., A. Sapate, A. Pipare, and V. Sonule. 2025. "SPI-based Drought Prediction in Maharashtra using LSTM: Analyzing Precipitation Trends from (2014 to 2024)." *JETIR*.

Min, L. 2025. "Hydrological Drought Prediction and its Influencing Factors Analysis based on a Machine Learning Model." *EGUsphere*.

Nandgude, N., T. P. Singh, S. Nandgude, and M. Tiwari. 2023. "Drought Prediction: A Comprehensive Review of Different Drought Prediction Models and Adopted Technologies." *Sustainability*.

Pande, C. B., et al. 2024. "Forecasting of Meteorological Drought using Ensemble and ML Models (Nashik District, Maharashtra)." *Environmental Sciences Europe*.

Park, H., K. Kim, and D. K. Lee. 2019. "Prediction of Severe Drought Area based on Random Forest: Using Satellite Image and Topography Data." *Water*.

Qaisrani, Z. N., A. Nusrat, J. Iqbal, et al. 2025. "Evaluating and Mapping Spatial Drought in Northeast Thailand: Utilizing Analytic Hierarchy Process and Random Forest Algorithms." *Geocarto International* 40 (1).

Rezaiy, R., and A. Shabri. 2025. "Integrating Wavelet Transform and Support Vector Machine for Improved Drought Forecasting based on Standardized Precipitation Index." *Journal of Hydroinformatics*.

Tihi, A., and V. Popov. 2023. "A Comparison of ARIMA and Random Forest Time Series Models for Predicting Drought Indicators." *Water*.

Watford, A., C. T. Bauch, and M. Anand. 2024. "Dynamical Systems-inspired Machine Learning Methods for Drought Prediction." *Ecological Informatics* 84.

Zeynolabedin, A., M. Ehteshami, and J. Tamošaitienė. 2022. "Application of Meteorological, Hydrological and Remote Sensing Data to Develop a Hybrid Index for Drought Assessment." *Hydrological Sciences Journal* 67 (5): 784–797.

Zhang, N., X. Dai, M. A. Ehsan, and T. Deksissa. n.d. "Development of a Drought Prediction System based on Long Short-term Memory Networks (LSTM)." *MERRA-Land Dataset Publication*.